

Light-Operated APPLIANCE SWITCH

SURESH CHANDRA DWIVEDI

Automation is now integral to daily life, with mobile phones serving as essential tools. This device employs a mobile phone torch to control domestic appliances, exemplifying an efficient light-operated switch. By responding to light intensity, it activates or deactivates appliances with ease.

The switch detects ambient light through a light-dependent resistor (LDR) or photodiode. A focused

beam from a mobile torch, standard torch, or laser torch triggers the system—powering the appliance on with the first focus and off with the second. The LDR in circuit translates changes in light intensity into electrical signals, driving the switching mechanism.

With versatile applications, this device can be handy for individuals with physical disabilities, enabling effortless appliance control. It is

PARTS LIST

Semiconductors:

IC1 (IC1A-IC1B)	- LM358 op-amp
IC2	- 4017 decade counter
T1	- BC337 NPN transistor
D1	- 1N4007 rectifier diode
LED1	- 5mm red LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 100k Ω
R2, R3	- 4.7k Ω
R4, R5	- 1k Ω

Capacitor:

C1	- 1 μ F, 25V electrolytic
----	-------------------------------

Miscellaneous:

RL1	- 12V, SPDT relay
CON1-CON3	- 2-pin connector
LDR1	- Small light-dependent resistor
	- 12V battery
	- Mobile phone/torch or laser light
	- Appliance (60W 230V AC bulb)

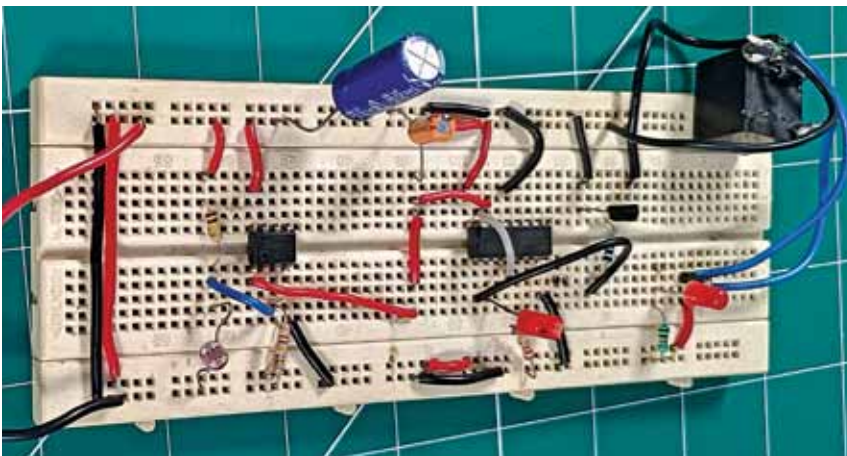


Fig. 1: Author's prototype

equally effective for managing indoor and outdoor lighting in homes. Fig. 1 shows the author's prototype on a breadboard.

Circuit and working

Fig. 2 shows the the light-operated circuit diagram to switch on/off appliances. The system is built around dual op-amp LM358 (IC1), decade counter IC 4017 (IC2), a 12V single-pole double-throw relay (RL1), and a few other components.

The working principle revolves around the LDR, whose resistance changes based on the intensity of light falling on it. The LDR exhibits high resistance in darkness or low-light conditions, reducing

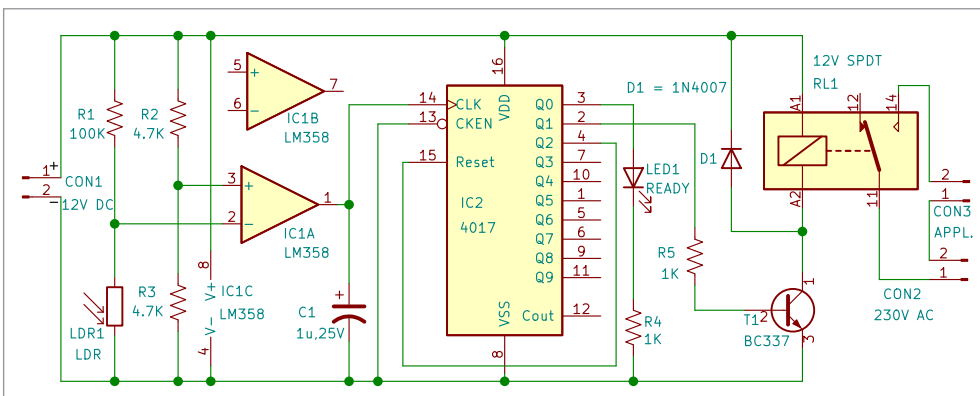


Fig. 2: Circuit diagram